$11010111.110 \rightarrow 1\times2^7+1\times2^6+1\times2^4+1\times2^2+1\times2^1+1\times2^0+1\times2^{-1}+1\times2^{-2}$
$=(215.\cancel{0}75)_{10}$

$\boxed{11}\boxed{010}\boxed{111}.\boxed{110} \rightarrow (327.6)_8$

3 2 7 6

$\boxed{1101}\boxed{0111}.\boxed{1100} \rightarrow (D7.C)_{16}$

D 7 C

c) octal 623.77 to decimal, binary and hexadecimal

$(623.77)_8 \rightarrow 6\times8^2+2\times8^1+3\times8^0+7\times8^{-1}+7\times8^{-2} = (403.984375)_{10}$

$\boxed{6}\boxed{2}\boxed{3}.\boxed{7}\boxed{7} \rightarrow (110010011.111111)_2$

110 010 011 111 111

$\boxed{1}\boxed{1001}\boxed{0011}.\boxed{1111}\boxed{1100}$

1 9 3 F C

$(193.FC)_{16}$ □

d) hexadecimal 2AC5.D to decimal, octal and binary

$(2AC5.D)_{16} \rightarrow 2\times16^3+10\times16^2+12\times16^1+5\times16^0+13\times16^{-1}$
$=(10949.8125)_{10}$

2 A C 5 . D $\rightarrow (0010101011000101.1101)_2$

$\boxed{0}\boxed{010}\boxed{101}\boxed{011}\boxed{000}\boxed{101}\ \boxed{110}\boxed{100}$

0 2 5 3 0 5 6 4

$(025305.64)_8$ □.

1-8. Obtain the 1's and 2's complement of the following binary numbers:
1010101, 0111000, 0000001, 10000, 00000.

a) 1's complement of 1010101 → 0101010
2's complement of 1010101 → 0101010

$$\begin{array}{r} 0101010 \\ +\quad 1 \\ \hline 0101011 \end{array}$$

b) 1's complement of 0111000 → 1000111
2's complement of 0111000 → 1000111

$$\begin{array}{r} 1000111 \\ +\quad 1 \\ \hline 1001000 \end{array}$$